

Principals of Embedded Systems - Project Proposal

STM32 I2C Multi-Device Integration Project

****Project Overview****

This project focuses on implementing I2C communication between an STM32 microcontroller and multiple I2C devices, starting with an SSD1306 OLED display and potentially expanding to include additional sensors.

****Primary Objectives****

1. Establish I2C communication between STM32 and SSD1306 OLED display
2. Successfully display text and basic graphics on the OLED screen
3. Implement proper I2C initialization and error handling
4. Create modular and reusable I2C driver code

****Extended Goals** (Time Permitting)**

- Integration of 2-3 additional I2C sensors into the network
- Implementation of multi-slave I2C architecture
- Development of an interrupt-based I2C communication system
- Creation of a simple user interface on the OLED display showing sensor data

****Technical Requirements****

- Hardware
 - * STM32 Microcontroller
 - * SSD1306 OLED Display (128x64 resolution)
 - * Additional I2C sensors (TBD)
 - * Required connecting wires and pull-up resistors
- Software
 - * STM32CubeIDE/STM32CubeMX
 - * I2C drivers
 - * SSD1306 display driver library

****Project Timeline****

Week 1-2: Basic I2C setup and OLED display integration

Week 3-4: Additional sensor integration and testing (if time permits)

****Key Learning Outcomes****

1. Understanding I2C protocol fundamentals
 - Master-slave communication
 - Address-based device selection
 - Clock synchronization
 - Error detection and handling
2. OLED Display Programming
 - Display initialization sequence
 - Buffer management
 - Graphics primitives implementation
3. Multi-device I2C Network (Extended)
 - Bus arbitration
 - Clock stretching
 - Multiple slave management

****Expected Deliverables****

1. Working prototype with OLED display
2. Well-documented source code
3. Basic demonstration program
4. Project documentation including setup and usage instructions

****Risk Assessment****

- I2C timing issues might require debugging
- Hardware compatibility concerns
- Time constraints for implementing extended goals